EXCELLENCE

CLASS 11 - 2026-27

Physics | Motion 05

Motion in a Straight Line

Acceleration · later intervals · when a stays constant

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What is new today

1. Average velocity on an interval that starts after $t = 0$
2. Which of two x–t curves is faster, early and late
3. Acceleration: $a = dv/dt$ and $a = v\,dv/dx$
4. Speed can rise while a falls; $v = 0$ can sit with $a \neq 0$
5. Opposite velocity and acceleration
6. A later velocity from two readings, when a is constant
7. Which formulas hide a constant acceleration

Average velocity between t = 2 s and t = 5 s

Position is $x = 5 + 2t^2$. The interval asked for starts at t = 2 s, so the average uses those two clock readings.

$x(5) = 5 + 2(25) = 55$ m

$x(2) = 5 + 2(4) = 13$ m

$v_{av}g = (55 - 13) / (5 - 2) = 42 / 3 = 14$ m/s

v = 4t stays positive from 2 s to 5 s, so the particle keeps the same direction. Distance equals the 42 m displacement, and the average speed is 14 m/s as well.

14 m/s

The integral agrees, with one careful step

Average velocity is also the time-average of v. Here v = dx/dt = 4t.

$$v_{avg} = (1 / 3) \int_2^5 4t\, dt = [2t^2]_2^5 / 3$$

$$= 2(5^2 - 2^2) / 3 = 2(25 - 4) / 3 = 14 \text{ m/s}$$

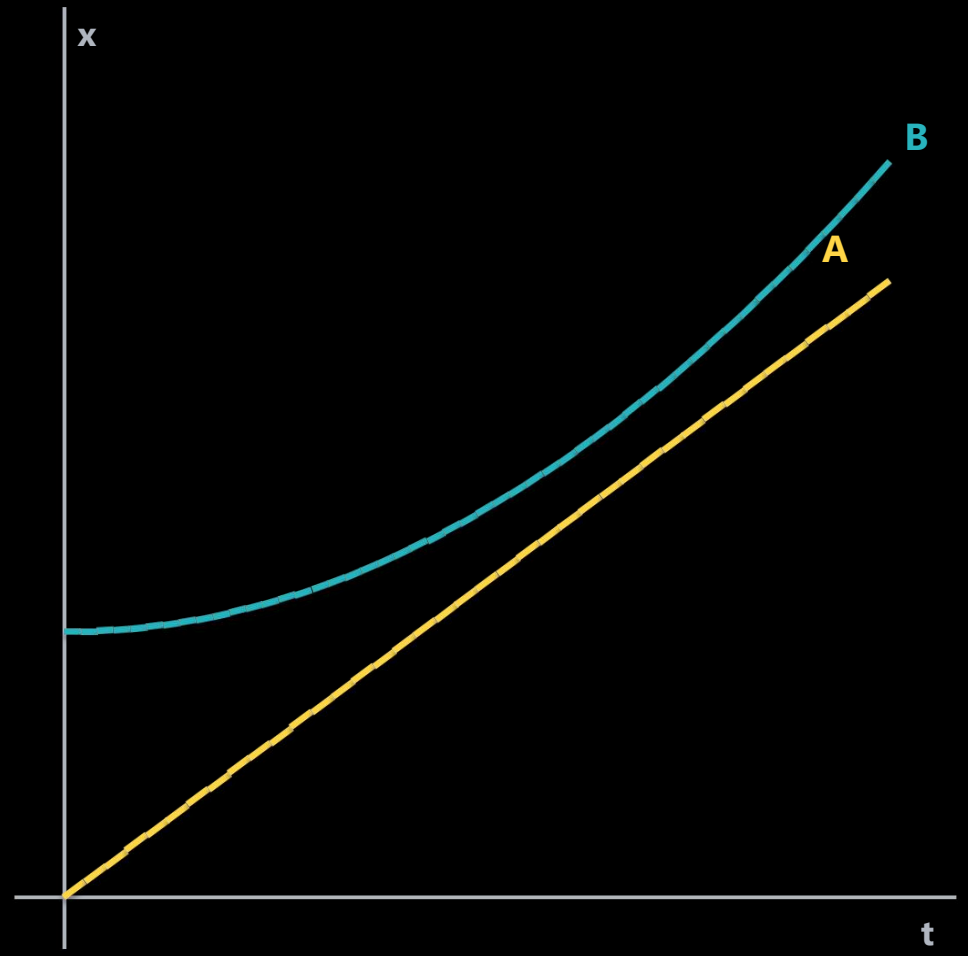
2(5 − 2) = 6 uses the × in place of their squares. The antiderivative of 4t is $2t^2$, so the squares stay.

Endpoints and the integral both give 14 m/s

Which particle is faster?

A and B move on the x–t graph. Which statement is correct?

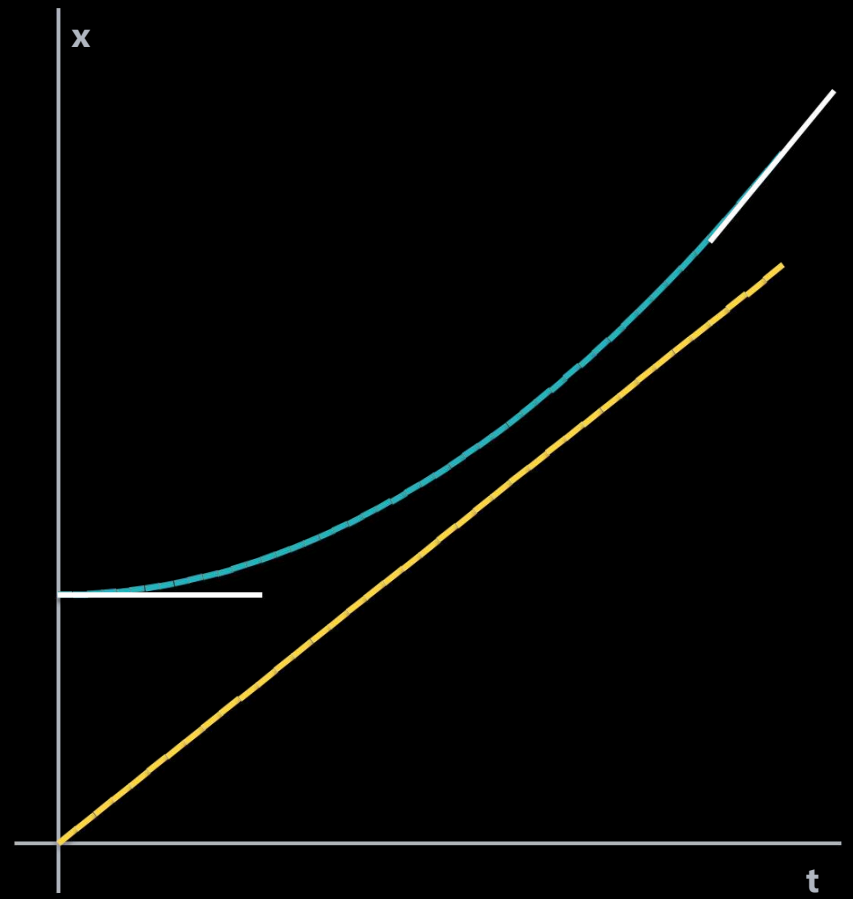
1. B is faster at the start and slower at the end
2. B is slower at the start and faster at the end
3. B is faster at the start and at the end
4. B is slower at the start and at the end



Solution: Which particle is faster?

The slope of an x–t graph is the velocity. A is a straight line, so its speed stays fixed. B opens with a horizontal tangent, so B's initial speed is zero, below A's. At the right-hand end B's tangent is steeper than A's line, so B is faster there.

Answer: B is slower at first, faster at the end



Rough work / extra work

Acceleration is how velocity changes

Velocity tells how fast the position is changing, and in which direction. Acceleration tells how fast that velocity itself is changing. Its direction is the direction of the change in velocity.

$$a_{av}g = (vf - v_i) / \Delta t$$

$$a = dv/dt = d^2x/dt^2$$

$$a = v\, dv/dx$$

Constant velocity means a = 0

Same clock, five different accelerations

Each row gives the velocity at t = 0, 1, 2 and 3 s. The acceleration column is the change from one second to the next.

Story	Velocities (m/s)	Accelerations (m/s^2)
Steady	10, 10, 10, 10	0, 0, 0
Even speed-up	10, 20, 30, 40	10, 10, 10
Gentler speed-up	10, 20, 28, 34	10, 8, 6
Harder speed-up	10, 20, 35, 55	10, 15, 20
Slowing down	20, 10, 6, 4	−10, −4, −2

On the gentler row the velocity keeps rising while the acceleration falls from 10 to 6. Rising velocity and falling acceleration can happen together.

Slowing down, with a growing acceleration

Velocity readings: 20, 18, 14, 8 m/s at t = 0, 1, 2, 3 s. The object is still moving forward, and it is moving more slowly each second.

$a = (18 - 20) / 1 = -2\ m/s^2$

$a = (14 - 18) / 1 = -4\ m/s^2$

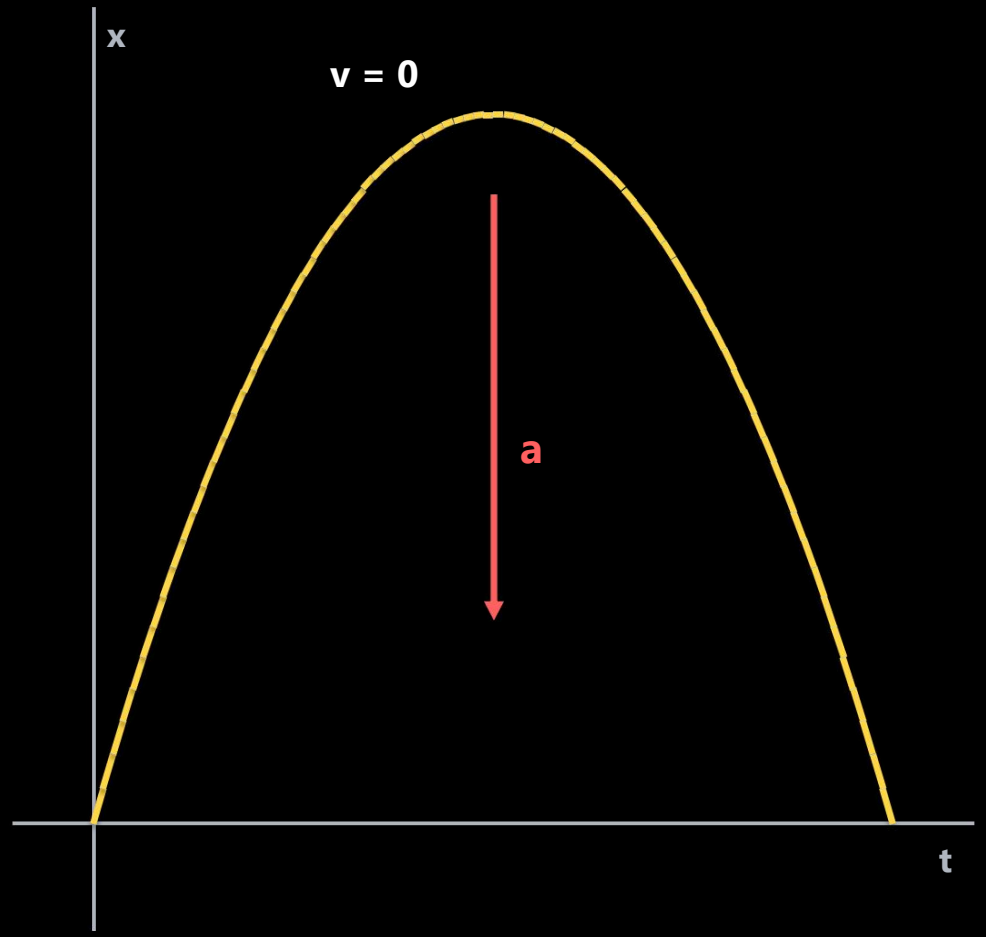
$a = (8 - 14) / 1 = -6\ m/s^2$

Velocity falls. The size of a grows from 2 to 6.

v = 0 while acceleration is still there

At the top of a vertical throw the velocity is zero for one instant. The acceleration is still g, downward. That acceleration carries the velocity from upward, through zero, and on to downward, so the object turns around.

v = 0 and a = g downward, together



Rough work / extra work

Which statements can be true?

Which option is correct?

1. Velocity may increase while acceleration decreases
2. Velocity may decrease while the size of the acceleration increases
3. Acceleration may be non-zero at the instant velocity is zero
4. All of these

Solution: Which statements can be true?

- Gentler speed-up: 10, 20, 28, 34 m/s with a = 10, 8, 6 m/s^2.
- Harder slowing: 20, 18, 14, 8 m/s with a = −2, −4, −6 m/s^2.
- Top of a throw: v = 0 while a = g downward.

Answer: all of these

Velocity and acceleration in opposite directions

Opposite directions mean the acceleration points against the velocity. That part of the acceleration reduces the speed. A perpendicular acceleration, as in uniform circular motion, leaves the speed unchanged. The opposite case is the one in which the speed falls.



Speed must be decreasing

Two readings with constant acceleration

Acceleration is 2 m/s^2 and the velocity at t = 0 is 10 m/s. Find the velocity at t = 4 s.

$v = 10 + 2 \times 4 = 18$ m/s

A second object goes from 10 m/s to 40 m/s in 6 s. The acceleration over that interval is the change in velocity divided by the time.

$a_{av}g = (40 - 10) / 6 = 5$ m/s^2

18 m/s and 5 m/s^2

A later velocity, once a is constant

Velocity is 20 m/s at t = 2 s and 32 m/s at t = 5 s. These two readings determine the velocity at t = 7 s when the acceleration is constant.

$a = (32 - 20) / (5 - 2) = 12 / 3 = 4\ m/s^2$

From t = 5 s to t = 7 s is another 2 s

$v(7) = 32 + 4 \times 2 = 40\ m/s$

40 m/s

Rough work / extra work

Acceleration from x(t)

Differentiate once for v, and once more for a.

$x = 4t^2 + 6 \rightarrow v = 8t \rightarrow a = 8\ m/s^2$

$x = 3t^2 + 4t + 6 \rightarrow v = 6t + 4 \rightarrow a = 6\ m/s^2$

$x = 2t^3 + 5t \rightarrow v = 6t^2 + 5 \rightarrow a = 12t\ m/s^2$

$x = t^4 + 4t \rightarrow v = 4t^3 + 4 \rightarrow a = 12t^2\ m/s^2$

Acceleration from v, including v(x)

$v = 3t^2 + 4 \quad \rightarrow \quad a = 6t \text{ m/s}^2$

$v = t^3 + 4 \quad \rightarrow \quad a = 3t^2 \text{ m/s}^2$

When velocity is given against position, differentiate with respect to x and multiply by v.

$v = 2x + 4 \quad dv/dx = 2$

$a = v\,dv/dx = (2x + 4) \times 2 = 4x + 8 \text{ m/s}^2$

One instant: position, velocity, acceleration

$x = 2t^3 - 4t^2 + 4$. Read all three at $t = 2$ s.

$x(2) = 2(8) - 4(4) + 4 = 4$ m

$v = 6t^2 - 8t \quad \rightarrow \quad v(2) = 24 - 16 = 8$ m/s

$a = 12t - 8 \quad \rightarrow \quad a(2) = 24 - 8 = 16$ m/s^2

4 m , 8 m/s , 16 m/s^2

The instant when acceleration is zero

For $x = t^3 - 6t^2 + 10$, set $a = 0$ and then read the velocity there.

$v = 3t^2 - 12t$

$a = 6t - 12 = 0 \quad \rightarrow \quad t = 2\ s$

$v(2) = 3(4) - 12(2) = -12\ m/s$

A second position, $x = t^3 - 9t^2 + 4$, has $a = 6t - 18$. That is zero at $t = 3$ s, and $v(3) = 3(9) - 18(3) = -27$ m/s.

$t = 2\ s, v = -12\ m/s$ and $t = 3\ s, v = -27\ m/s$

Acceleration at the instant velocity is zero

$x = 2t^3 - 24t + 4$. Find a at every time where $v = 0$.

$v = 6t^2 - 24 = 6(t^2 - 4) = 0$

$t = 2$ s and $t = -2$ s

$a = 12t$

$a(2) = 24$ m/s^2 $a(-2) = -24$ m/s^2

24 m/s^2 at $t = 2$ s, and -24 m/s^2 at $t = -2$ s

Rough work / extra work

Six formulas whose acceleration is a pure number

In this list y is a constant. Differentiate until a appears. These six come out as numbers.

- $\sqrt{x} = t - 3 \quad \rightarrow \quad x = (t - 3)^2 \quad \rightarrow \quad a = 2\ m/s^2$
- $x = t^2 + y \quad \rightarrow \quad a = 2\ m/s^2$
- $x = 4t^2 + 6 \quad \rightarrow \quad a = 8\ m/s^2$
- $v = 6t - y \quad \rightarrow \quad a = 6\ m/s^2$
- $v = \sqrt{x} \quad \rightarrow \quad a = 1/2\ m/s^2$
- $a = 5\ m/s^2$ is already constant

$v = \sqrt{x}$ really is constant acceleration

$v = \sqrt{x}$

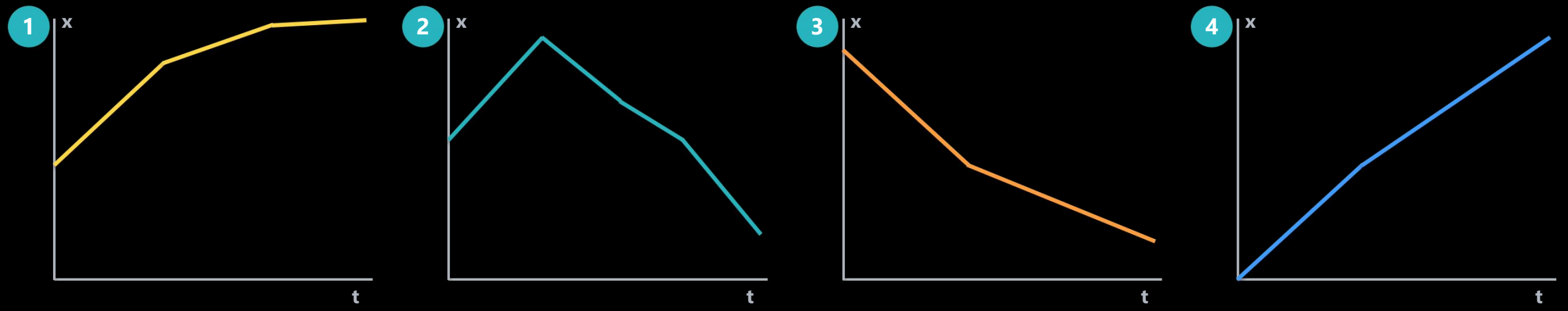
$dv/dx = 1 / (2\sqrt{x})$

$a = v\,dv/dx = \sqrt{x} \times 1/(2\sqrt{x}) = 1/2\ m/s^2$

Three neighbours still depend on t or on x, so their acceleration keeps changing: $v = 3t^2 + 4$ gives $a = 6t$; $v = 2x + 3$ gives $a = 4x + 6$; $x = A\sin(\omega t)$ gives $a = -A\omega^2 \sin(\omega t)$.

Which graph can have average velocity zero?

Average velocity from t = 0 to some later time can be zero for only one of these graphs. Which one?



Solution: Which graph can have average velocity zero?

Average velocity from t = 0 to a later time is zero when the position has returned to its starting value. Graph 2 climbs and then crosses that same height again, so there is a later time with x equal to the start. Graphs 1, 3 and 4 leave their starting height and remain on one side of it.

Answer: graph 2

Only today’s lines

- Between two times, average velocity is the change in position divided by the change in time.
- $a = dv/dt = d^2x/dt^2$, and $a = v\,dv/dx$.
- Constant velocity means $a = 0$. At the top of a throw, $v = 0$ while a is still g downward.
- When a points with v the speed rises. When a points against v the speed falls.
- Two velocity readings fix a later velocity once a is constant: $v = u + at$.
- a is constant when it comes out as a pure number. For $v = \sqrt{x}$, that number is $1/2$.

Rough work / extra work

Thank you

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